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Optimization of surface finish in fused deposition modeled components using hybrid metaheuristic techniques and abrasive flow machining

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Abstract

Fused deposition Modelling (FDM) technique is extensively used in industrial applications. The main advantage of this technique is to provide flexibility and ease of operation, but at the same time, surface roughness is a significant drawback to FDM, caused by layer-by-layer deposition resulting in staircase effects and surface irregularities. For this purpose, the present study used Abrasive Flow Machining (AFM) to enhance the surface quality of PLA/Cu composite specimens produced by FDM with a biodegradable abrasive medium made from paper pulp, peanut oil, and stainless-steel particles. Experimental investigations were carried out to analyze the effect of the number of strokes, abrasive concentration, and extrusion pressure on surface roughness. Taguchi L9 experimental data were used to construct a predictive surface roughness model and then combined with a hybrid optimization algorithm, namely Cuckoo Search-Differential Evolution (CS-DE), to determine the optimum parameters of the AFM process. The optimum parameter combination was determined as 100 strokes, 0.00981 MPa extrusion pressure, and 35 wt% abrasive concentrations from the optimization results. In such a situation, the surface roughness predicted was 6.193624 μm , and the confirmation experiment result was 6.30 μm with a prediction error of about 1.71%. The SEM-based surface characterization results confirmed a significant decrease in layer-induced asperities and ridge height, which confirmed an effective surface smoothing process. The results show that the proposed hybrid CS-DE optimization approach offers an effective and reliable methodology to improve the surface finish of the FDM-fabricated parts treated with AFM, and the agreement between the proposed and experimental results is excellent.

1 Introduction

FDM is considered to be one of the most widely adopted additive manufacturing technologies [1] due to its operational simplicity, cost-effectiveness, and capability to produce geometrically complex parts [2, 3]. The process is carried out by extracting the



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thermoplastic filament through a heated nozzle, where it is dispersed over the surface and converted into the desired shape. Despite all these advantages or benefits still some major concern like surface roughness, which affects both performance and aesthetic characteristics of manufactured components [4, 5].

The value of Surface roughness in FDM parts depends upon multiple things like the layer-by-layer deposition process, staircase effect on curved surfaces as shown in Fig. 1, material extrusion characteristics, and various process parameters [3, 4]. The optimization of these parameters is critical for minimizing surface roughness while maintaining dimensional accuracy and mechanical properties. traditional approach of optimization depends upon the hit-and-trial method and design of experiments, which is not suitable for the complex and multi-level parameter approach. Thereby, Metaheuristic optimization algorithms have emerged as powerful tools for addressing complex, multi-parameter optimization problems in manufacturing [1, 2, 6–8]. Apart from this, hybrid approaches that combine multiple algorithmic strategies show highly effective of simplify and providing strength to the mechanism adopted for optimization. Specifically, hybrid algorithms integrating Cuckoo Search (CS) and Differential Evolution (DE) have shown promise in balancing globally and locally accepted techniques for addressing premature convergence and enhancing population diversity [1, 9–12].

Post-processing techniques such as AFM offer significant potential for enhancing the surface quality of FDM parts [3, 13–19]. AFM is a non-traditional method where viscoelastic fluid is passed with extruded pressure through a small opening, and thereby internal and external improved surface finish is achieved by micro-cutting action and rubbing of abrasive particles against each other. The optimization of AFM parameters, including the number of cycles(stroke), abrasive concentration(slurry), and extrusion pressure, is essential for achieving desired surface characteristics.

Enhanced dimensional accuracy and surface integrity of parts produced from conventional and non-conventional machining processes are the focus of recent advances in advanced manufacturing. Electrical Discharge Machining (EDM), Micro-EDM (μ -EDM), and Wire Electrical Discharge Machining (WEDM) have gained a lot of interest because these can machine complex geometries and difficult materials [20]. Recently, it has been shown that the dielectric media have a significant effect on the efficiency of the machining process, tool wear, surface morphology, and re-cut layer formation. However, despite these advances, there is still a need for better quality surface finish in polymer components fabricated by additive manufacturing processes, which requires the identification and application of appropriate post-processing and optimization methods.

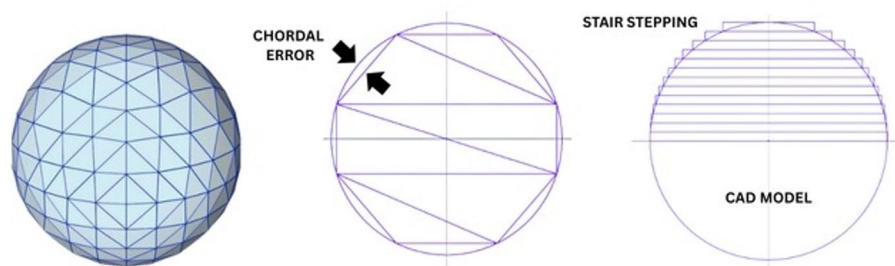


Fig. 1 Different errors associated with the FDM process, namely truncated, chordal, and stair stepping

The current research is focused on hybrid metaheuristic optimization for FDM surface roughness minimization, with emphasis on CS-DE hybrid algorithms and AFM post-processing. The main objective of the research is to provide the optimal solution by using a hybrid mathematical model for improving the surface finish of 3d printed parts made by FDM and processed by the AFM process using biodegradable media so that it can be used in various medical and daily practice applications. There are some studies on the parameters of FDM with abrasive material, like abrasive size, media viscosity, and flow rate, using a novel eco-friendly abrasive media [21]. This study focuses on other parameter which are also play a significant role but are very rarely considered, such as the parameters like no. of cycles, which directly involve the material removal rate (MRR) and surface finish. So it's important to consider these parameters and find out the relation along with the impact of these parameters on surface finish and optimize these parameters by using a hybrid algorithm.

1.1 Background and theoretical foundations

1.1.1 Fused deposition modelling and surface quality challenges

FDM is the process which is done by using thermoplastic material heated and extruded through a nozzle and poured onto the surface layer by layer to make printed parts. As the FDM parts are made layer by layer, so it contain or contribute to roughness, orientation marks or the surface, staircase effects on inclined and curved surfaces, and inter-layer bonding irregularities [4, 5], as shown in Fig. 1.

Surface roughness of FDM-manufactured parts depends on plenty of factors. Vahabli and Rahmati [4] did research on a hybrid estimation approach combining analytical Modelling and empirical investigation to understand the effect of surface roughness on FDM parts. It tried to demonstrate the complexity of roughness prediction and make it easier to apply [4]. The work showcases the surface roughness that varies significantly across different surface orientations and geometries of the same workpiece.

The advanced characterization of the techniques is mostly used for FDM parts for their quality and structural orientation. Computed tomography analysis shows dimension precision, deviation in the parts made up by FDM techniques, and their geometry [22]. These dimensional variations, coupled with surface roughness, require a comprehensive optimization strategy and the multiple quality attributes that play important roles simultaneously [23].

1.1.2 Metaheuristic optimization algorithms

Metaheuristic algorithms are high-level problem-independent algorithmic whom frameworks provide sufficiently good solutions to optimization problems, particularly when classical optimization methods are impractical because of complexity, non-linearity, or computational constraints [24]. These algorithms are inspired by natural phenomena, such as physical processes or social behaviours, and have been successfully applied to manufacturing optimization problems.

Cuckoo Search (CS) is a nature-inspired metaheuristic algorithm based on the obligate brood parasitism of some cuckoo species combined with Levy flight behavior of certain birds and fruit flies [1, 12, 25]. CS employs a combination of local random walks and global explorative jumps, making it effective for exploring large search spaces. The

algorithm has been successfully applied to various manufacturing optimization problems, including surface roughness optimization in machining operations [26].

Differential Evolution (DE) is a population-based stochastic optimization algorithm that works by mutation, crossover, and selection operations [1, 27]. DE is particularly effective for continuous optimization problems and provides local search capability. Population diversity is maintained by the algorithm through its mutation strategy, which generates trial vectors by combining randomly selected population members [27].

Hybrid Metaheuristic Approaches combine multiple algorithm approach will provide the strength by mitigating or reducing the weakness of each algorithm alone. There are different hybrid algorithm which have their own strength and weaknesses; some are acceptable globally, or some are best locally [1, 11, 12]. Hybrid approaches aim to achieve balanced and overall optimized performance.

There are plenty of other metaheuristic algorithms also applied to FDM optimization, namely as Particle Swarm Optimization (PSO), Genetic Algorithms (GA), Teaching-Learning Based Optimization (TLBO), Grey Wolf Optimizer (GWO), and Dragonfly Algorithm (DA) [2, 6, 22, 28]. Comparative study of these algorithms is different and relatively different for different situations and objectives [24].

Biswas et al. [29] performed experimental study on enhancing the prediction and optimization of Wire Electrical Discharge Machining (WEDM) for titanium alloy Grade 5 (Ti-6Al-4 V) using a hybrid approach combining Artificial Neural Networks (ANN) with meta-heuristic algorithms. The authors systematically analyzed key WEDM process parameters—such as pulse on time, pulse off time, servo voltage, and wire feed rate—and their effects on critical output responses including material removal rate, surface roughness, and wire wear ratio. An ANN model was developed for accurate prediction of machining outcomes, and meta-heuristic optimization techniques were applied to identify optimal parametric combinations, resulting in significantly improved machining performance and surface quality for this difficult-to-machine aerospace alloy.

Table 1 shows the latest research related to surface finish parameter optimization. In the given table, it is found that no optimal solutions are to be formed for different conditions and different parameters. Many research uses various optimization and integrated optimization and meta-heuristic methods, but contains limitations for other parameters. So, from the above literature review table and the research gap, it's significant to employ the Hybrid Metaheuristic Techniques (CS + DE), which overcome and provide balanced global exploration through Levy flight-based random search, helping the optimization process escape local optima and local exploitation through mutation and crossover operations, refining promising solutions.

2 Materials and methods

FDM is a widely used technique to manufacture different types of parts depending on their applications. In the FDM process, generally thermoplastic materials are used, which form any shape on the application of heating, due to which the part manufactured through this technique is lightweight and easy to form. However, on the other hand, the same reason becomes the drawback for this process, as the materials used for the process are PLA, ABS, Nylon, PEEK, TBU, etc. These materials have less tensile strength and toughness to make the process more suitable for wide application. Different metal materials are reinforced into the thermoplastic matrix material. In the present study,

Table 1 Literature review on various processes used to improve surface quality

Sr no.	Process used by authors	Optimization technique used	Objective	Finding	Research gap
1	AFM [30]	NA	To improve the surface quality	AFM significantly improves the surface finish	Validation of data and no optimization technique used
2	Additive manufacturing post processing [31]	Comparative review study	To reduce surface roughness	Summarized advanced finishing methods, including fluid-assisted and abrasive finishing processes for AM components	Does not address hybrid metaheuristic optimization.
3	FDM + AFM [32]	Taguchi L9	To improve the surface finish	The developed abrasive media was found to be effective for finishing FDM printed parts using AFM	No biodegradable material are used
4	FDM + AFM [33]	Parametric Study	Surface roughness reduction	AFM effectively improved the AM surface quality	No metaheuristic optimization applied
5	EDM machine parameter [23]	Taguchi L27 OA	Parameters optimization	Develop	integrated neural network and meta-heuristic method
6	FDM+PA12-CF Parts [34]	Integrated Response Surface Methodology, Grey Relational Analysis, and Grey Wolf Optimization	The primary objective of this research is to investigate the relationship between eight key parameters in fused deposition Modeling (FDM)	integrated approach combining RSM, GRA, and GWO can effectively optimize FDM process parameters for PA12-CF parts, leading to superior mechanical properties and surface finish, with significant benefits for industrial applications.	Need for Integrated Optimization Methodology
7	FDM +PETG and ASA Parts [35]	Multi-Response Optimization	To optimize the parameters of FDM, Value Maximization, and Identification of Optimal Parameters	Ultimately, a universal set of optimal parameters, $L_h = 0.20$ mm and $Id = 100\%$, was identified for efficient FDM manufacturing of both PETG and ASA parts.	The ultimate research gap is to determine the optimal FDM parameters that achieve the best balance between mechanical performance and production cost for PETG and ASA parts.
8	FDM	Taguchi L18 OA PSO-BFO evolutionary algorithm	To enhance the performance of FDM-manufactured parts by optimizing key process parameters using a hybrid PSO-BFO algorithm, targeting improved mechanical properties and surface quality.	FDM parameters are adjusted to achieve desired performance levels. Hybrid PSO-BFO Algorithm	Challenge in Parameter Selection

Table 1 (continued)

Sr no.	Process used by authors	Optimization technique used	Objective	Finding	Research gap
9	micro-electrical discharge(μ -ED)	NA	Effect of vegetable oil as an eco-friendly dielectric in μ -ED nitinol alloy enhanced bio-compatibility	This study underscores the potential of vegetable oils in promoting environmentally friendly manufacturing practices while achieving high-quality outcomes for medical-grade Nitinol components.	No meta-heuristic optimization applied
10	Computational Algorithm Implementation		statistical validation through 100 independent runs and multiple benchmark functions to ensure the robustness of conclusions	Enhances understanding of surface finishing processes through combined experimental-numerical investigation of AFM.	Limited research on hybrid metaheuristic approaches combining strengths of multiple algorithms
11	Wire Electrical Discharge Machining (WEDM) for titanium alloy Grade 5 (Ti-6Al-4 V)	Hybrid approach combining Artificial Neural Networks (ANN) with meta-heuristic algorithms	Systematically analyzed key WEDM process parameters and their effects on critical output responses	An ANN model was developed for accurate prediction of machining outcomes, and meta-heuristic optimization techniques were applied to identify optimal parametric combinations, resulting in significantly improved machining performance and surface quality for this difficult-to-machine aerospace alloy.	NA

PLA and copper matrix are used to form the specimen, which is suitable for various medical and industrial applications. Reason for selecting this matrix as PLA and Copper both are bio compatible, biodegradable, and non-toxic in nature. To maintain the same properties, the same type of abrasive material is required, thereby paper pulp, vegetable oil, peanut oil, and stainless steel 316 L grade slurry material is used.

2.1 Abrasive flow machining for surface enhancement

AFM operates through the controlled flow of a viscoelastic abrasive medium across or through workpiece surfaces under pressure [13–19]. The abrasive media generally contain abrasive particles, which are blended with polymer based carries, so they also contain viscous and elastic properties, as shown in Fig. 2. This unique Rheological pattern will help to develop and maintain the complex geometric structure without compromising the stiffness, and it also helps in abrasive action. In this study, vertical AFM is used with the specifications as shown in Table 2.

The material removal during AFM is done by the abrasive particle and its micro-cutting action, which flows transverse to the workpiece and smooths the surface, as shown in Fig. 2. Shao et al. [18], working on the integrating model of AFM, which deals with the micro-cutting mechanism and surface generation model, provide detailed insights into the material removal process at the micro level. The work explained that MRR is mainly due to plastic deformation and microchip generation, which is integrated with the effect of abrasive particle geometry, their properties, and other process conditions, etc.

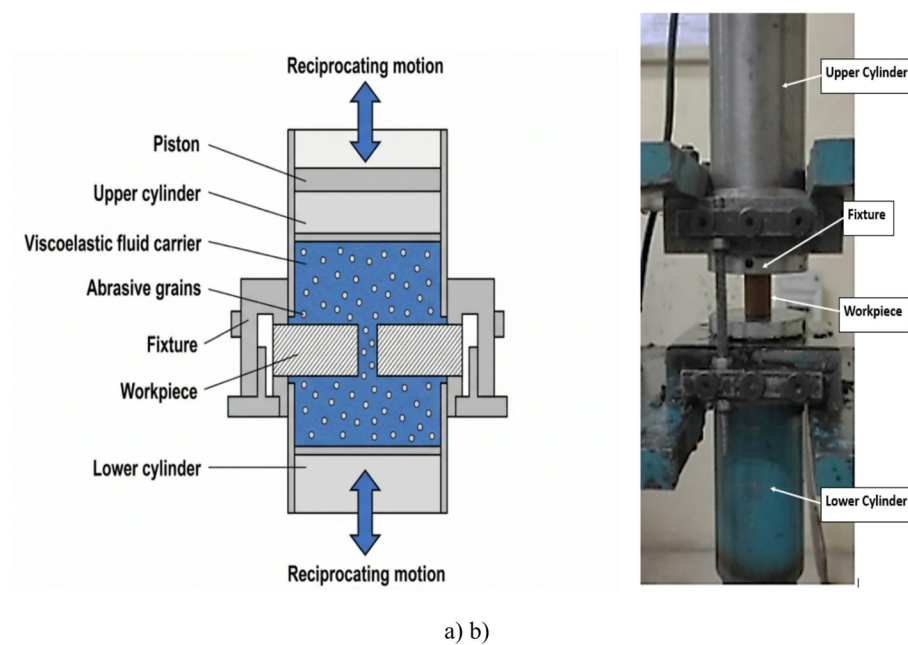


Fig. 2 **a** Fundamental principle of AFM with the main component used for processing; and **b** AFM machine setup with specimen

Table 2 Specification of vertical AFM

Sr no	Specification	Description	Units
1	Maximum Extrusion Pressure	10	MPa
2	Material of Hydraulic Cylinder	EN8	NA
3	Material of Piston	Grey Cast Iron	NA
4	Diameter of Hydraulic Cylinder	95	mm
5	Diameter of Piston	90	mm
6	Media Flow Volume per Pass	0.0003	m ³
7	Stroke Length	250	mm
8	Pressure Gauges	2 Nos	(each 3000 psi)
9	Pump	Hydraulic Gear Pump	NA
10	Motor	Flange type Motor	(2 H.P. and 1440 rpm)

Investigators Wang et al. [36] explored the structural changes of the surface texture as being transformed by the operation. It denotes that changes in the morphology of the surface occur through a gradual increase in its fineness. In the beginning, mainly larger irregularities are removed, and the surface valleys and tips are leveled. Later, smaller surface imperfections are erased one after another, leading to a near-mirror finish. The process of surface enhancement slows down with each successive cycle; thus, a determination of an optimal mode of treatment is implied, beyond which little can be gained by further cycles.

2.2 Key Process Parameters

There are many performance key parameters for the AFM processed FDM parts, which play a significant role in the surface outcome and texture of the surface, so these must be optimized to get the desired result. Table 3 indicates the detailed factor and level selected for the AFM Process to improve the surface quality. On the detailed study and research, there are key parameters that include:

Table 3 Factors and levels for parameters used in AFM machining operations

Factor notation	Component	Unit	Level 1	Level 2	Level 3
A	Stroke (no. of cycles)	Dimensionless	50	100	150
B	Extrusion Pressure	MPa	0.00981	0.0490	0.0980665
C	Abrasive concentration	%wt	20	35	50

Number of Cycles (or Strokes) represents the count of abrasive material moved from one end to the other end of the workpiece during the AFM process [13–15, 17]. Generally, increasing the number of cycles improves the surface finish with this process. However, the returns will diminish, and too many cycles will result in over-polishing, dimensional change, or the creation of undesirable surface characteristics. Optimization studies have pinpointed the maximum cycle numbers that effectively increase surface quality without sacrificing processing efficiency [15, 17].

Abrasive Concentration is the number of abrasive particles that are present in the totality of the flowing medium [14, 15, 17]. Abrasive concentration has a direct relation with the material removal rate. The higher the concentration, the higher the material removal rate, but it may also increase surface roughness if the concentration becomes excessive. Lower values of concentration may result in lowering the material removal rate. The optimum value of abrasive concentration depends upon the material, the type of finish required, and other process parameters [15, 17].

Extrusion Pressure controls the force with which the abrasive medium is pressed against the workpiece surface [14, 15, 17, 21]. At higher pressures, materials are destroyed rapidly, and, due to more frequent contact of abrasive particles with the work surface, a better surface finish can also be attained. Still, the application of very high pressure can result in the deformation of the workpiece, especially if it is a thin-walled or compliant structure, which are typical of FDM parts. Kumar et al. [17] tested the surface roughness improvement as a function of extrusion pressure, along with other AFM parameters [17].

Algorithm Flow.

- Initialization:** Randomly sets the initial population within the given range for Stroke, Pressure, and Abrasive Concentration.
- Initial Evaluation:** For each individual in the starting population, the fitness (predicted roughness) is assessed by the MLP Regressor. The first best is marked.
- Iteration Cycle:** Starts a loop of a set number of iterations. Each generation, the population is divided into two exclusive subsets according to a previously defined ratio (cs_ratio).
- Cuckoo Search (CS) Phase:** (i) A Levy flight random walk is applied to generate new candidate solutions for the selected CS individuals, directing the search toward the current best solution identified during the optimization process. (ii) The newly generated solution is then evaluated. If its fitness value is better than that of the previous solution, it replaces the previous solution. (iii) An “alien egg discovery” mechanism is applied, in which, with a probability of p.a., the current individual is discarded and replaced with a new randomly generated solution within the specified search bounds.
- Phase Differential Evolution (DE):** (i) For everyone else, creates a new vector by picking three different individuals from the population at random and creating a new mutant vector using these. (ii) Carries out the binomial crossover operation to

generate a trial vector from both the target and the mutant vector. (iii) Carries out a test on the trial vector. It is passed on to the next generation if it produces a less rough vector than the desired vector.

- f. **Global Update:** If the best solution found in the current generation is a better fit than the overall best solution found, it replaces the old solution.
- i. **Termination:** The loop will repeat until the maximum number of iterations has been reached. The algorithm provides the optimized final parameters and minimal roughness.

Mutation Strategy: The script uses 2 different mutation strategies, one for each sub-population:

- (i) **This search mutation is given the name Cuckoo Search Mutation (Levy Flight):**
The Cuckoo Search subset creates mutated positions by taking random walks on a Levy distribution. This is done in two steps: determining the bounds of the step size (with Mantegna's algorithm) and applying the position update.

Part A: Step Size Scaling Factor (σ).

$$\sigma = \left(\frac{G(1 + \beta) \cdot \sin\left(\frac{\pi\beta}{2}\right)}{G\left(\frac{1+\beta}{2}\right) \cdot \beta \cdot 2^{\frac{\beta-1}{2}}} \right) \quad (1)$$

Where Γ : The Gamma function, a mathematical extension of the factorial function to real and complex numbers.

Part B: Position Update Equation.

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \left(\frac{\mathbf{u} \cdot \mathbf{s}}{|\mathbf{v}|^{\frac{1}{\beta}}} \right) \cdot (\mathbf{x}_i - \mathbf{x}_{\text{best}}) \cdot \mathbf{N}(0,1) \quad (2)$$

Where $\mathbf{x}_{\text{new}}$ is the newly mutated parameter vector (the trial solution). $\mathbf{x}_i$ denotes the current parameter vector of the i th individual being mutated. and $\mathbf{v}$ are the random numerical values drawn from a standard normal distribution (mean of 0, variance of 1).

$\frac{\mathbf{u} \cdot \mathbf{s}}{|\mathbf{v}|^{\frac{1}{\beta}}}$ is the completed Levy flight step length calculation. $\mathbf{x}_{\text{best}}$ represents the best parameter combination for the experimental conditions considered in this study parameter vector discovered by any individual in the population up to the current generation.

$\mathbf{N}(0,1)$ is a vector of random noise drawn from a standard normal distribution, applied to each dimension to ensure the mutation explores multi-directionally rather than moving in a straight line toward or away from $\mathbf{x}_{\text{best}}$. The DE subset utilizes the classic.

- (ii) **DE/rand/1 mutation strategy.** For a given target vector, it selects three distinct random vectors ($\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$) from the current population and creates a mutant vector ($\mathbf{V}$) using the scaling factor (F):

$$V = a + F \cdot (b - c) \quad (3)$$

Where V is the generated mutant vector; a , b , and c are the three entirely distinct parameter vectors selected at random from the current generation's population. Crucially, these three must be mutually exclusive and also distinct from the current target vector being evaluated. F is the scaling factor (or differential weight). It is a predetermined scalar value that controls the magnitude of the difference vector, preventing the mutation from overshooting or stagnating. $b - c$ is the difference vector. It dictates both the direction and the baseline distance of the mutation step.

The mutant vector is then recombined with the target vector during the crossover step.

(iii) Stopping Criteria.

The algorithm has a fixed termination criterion according to the number of generations. It only stops execution when the generation counter hits the `max_iteration` value (default 150). The loop does not have any early stopping or tolerance limit (e.g., stop after 400 generations if the population hasn't converged yet); it will do all 400 iterations regardless of the convergence of the population.

The AFM process involves extruding an abrasive medium against the wall of the surface of the workpiece under controlled pressure. The abrasive particle embedded in the viscoelastic material removes the extra materials or chips by means of micro-cutting action or by a micro-plow mechanism [18]. There are plenty of parameters that play an important role in material removal rate, and hence the resulting surface characteristics depend on multiple process parameters, which include parameters like abrasive concentration, particle size and type, medium viscosity or consistency, extrusion pressure, number of cycles or strokes, and flow velocity [13–15, 17].

Wang et al. [13] investigated and found that the surface roughness of the workpiece depends upon the parameters that are selected for operations [13]. Their work demonstrates that the surface finish depends upon the surface material removal rates in the number of cycles; it can be improved by means of controlling these parameters, so it's important to determine the optimal number of cycles.

Dhull et al. [18] performed an experimental study on computational and mathematical analysis of hybrid AFM processes; it developed models that relate process parameters to surface quality outcomes [14]. Also, try to explain the importance of mathematical Modelling for predicting AFM performance and optimizing process parameters. Similarly, Shao et al. [18] conducted experimentation on integrated models of micro-cutting mechanics with precision surface generation in AFM and improved the theoretical understanding of material removal mechanisms.

The optimization of AFM process parameters has been addressed through various approaches. Gupta et al. [15] examined the effect and optimization of process parameters on surface roughness in AFM, they identified critical parameters and their interactions. Kumar et al. [17] focused on optimizing process parameters for effective material surface roughness improvement, demonstrating that systematic parameter optimization can significantly enhance AFM performance. Hashmi et al. [19] worked on a comprehensive computational study and performed simulations on the parameters of AFM.

The process of optimization of AFM was carried out using the Crow Search–Differential Evolution (CS–DE) optimization algorithm. The data acquired from the Taguchi L9 orthogonal array was used as the input data for optimization. The CS–DE algorithm used the response values that were measured in the experiment to assess the fitness of each solution and find the optimum set of process parameters. The optimization process was performed directly on the experimental Taguchi data without the use of an objective function built out of a regression model. The nine experimental results obtained from the Taguchi L9 design were first used to train a predictive Artificial Neural Network (ANN) model based on a Multi-Layer Perceptron (MLP). This initial model was subsequently used to generate additional synthetic data points within the experimental design space, thereby expanding the training datasets. A final ANN model was then trained using the expanded datasets. During optimization, the objective function corresponded to the ANN-predicted surface roughness. Each candidate solution generated by the CS–DE algorithm was supplied to the trained ANN model, which predicted the corresponding surface roughness value. These predicted values were used as the fitness function during the optimization process.

Figure 3 presents the overall experimental methodology employed in this study. Initially, a PLA–Cu composite filament was fabricated using a single-extruder machine for use in the FDM process. The tubular specimens were subsequently fabricated using a Creality Ender-3 3D printer with a build volume of $0.220 \times 0.220 \times 0.250 \text{ m}^3$. The other important specifications of the printer, such as the diameter is 1.75 mm, layer thickness range varying from 100 to 400 microns. The given filament is printed at ASTM standard D638. Then the parts are finished by a vertical AFM machine with specifications as mentioned in Table 2. In this vertical AFM machine, biodegradable material made up of Paper pulp + Peanut oil + SS316L is used to improve the application of the printed part. The stainless steel 316 L grade 325 mesh is used as it has biodegradable and bio-compatible (non-toxic) properties. To measure the surface finish of the specimen after post-processing, a Talysurf surface roughness tester is used as shown in Fig. 7e. Surface roughness measurements were recorded at three different locations on the specimen surface, preferably at the middle of the tubular specimen, as the maximum slurry will flow there, and the average value was used for analysis to minimize measurement uncertainty and improve repeatability.

2.3 Algorithmic architecture and integration strategies

Hybrid algorithms combining Cuckoo Search and Differential Evolution employ for enchaining the both algorithms' strengths and strategies by complementing each

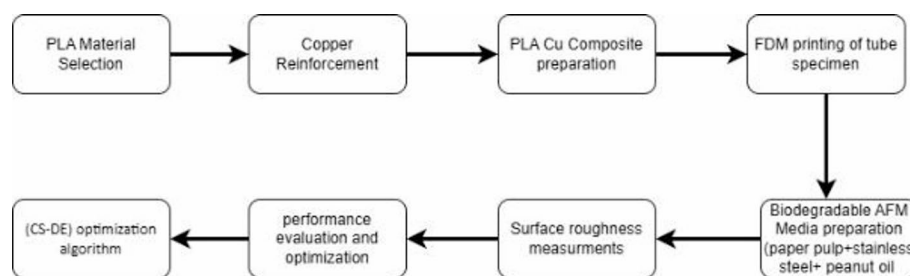


Fig. 3 Complete flow chart for planning and execution of experiment for surface finish through AFM technique and finding the optimizing parameter

other's weaknesses. The main concern of this hybrid model is to The Cuckoo Search (CS) algorithm enhances global exploration and improves the stability of the optimization process, whereas DE will provide the strengthen the search capability and improves population diversity, thereby reducing the likelihood of premature convergence, which play important roles in nonlinear models [1, 9–12].

Xi et al. [1] worked on a CSDE hybrid algorithm that introduces a control factor in the Levy Flight Random Walk (LFRW) component to determine when to apply a differential strategy for population diversity enhancement adds a control parameter in the LFRW module to switch between different strategies of differential strategy strengthening for enhancing population diversity. In the BSRW phase, it uses crossover and selection operators from DE with a self-adaptive crossover rate. This integration strategy enables the algorithm to adaptively balance exploration and exploitation according to search progress. The CSDE algorithm outperformed the original CS [5], DE, and PSO, and other hybrid variants on a larger set of benchmark functions [1]. Table 4 shows the complete parameter setting used to run the algorithm for the hybrid CS-DE optimization method.

Zhang et al. [12] focused on the improvement of the Cuckoo Search algorithm for differential inertial weight and evolution for the functional optimization process. Also, a differential inertia weight is considered to control the step size for better controlling and enhancing the approach by transitioning between exploration and exploitation phases. This approach will lead to promoting the integration of DE in mutation for maintaining diversification and controlling local optimization.

There are a few more strategies discussed in the literature; a few of them are classified into sub-branches, and each branch has unique treatment in different group, which permit the simultaneous exploration and exploitation in various zones of the search area [9, 10]. Other techniques activate the DE operation as the condition meets and trigger for convergence when the diversified population is fall below from the limit value [1, 11].

The popular method CS-DE hybrids found in the available literature where it have common designations, including CSDE, DECS, DECSKH (when Krill Herd optimization is also incorporated), and WCSDE (when inertia weight strategies are added) [1, 9–12]. Although all contain different names and nomenclature, the working principles are the same for each mechanism. CS is considered a global search framework; on the other

Table 4 Parameter Settings to run the algorithm for the hybrid CS-DE optimization method

Parameter	Variable Name	Value	Description
Population Size	pop_size	100	The total number of candidate solutions maintained per generation.
Iteration Count	max_iter	150	The maximum number of generations the algorithm will run.
Algorithm Split Ratio	cs_ratio	0.5	50% of the population undergoes Cuckoo Search; 50% undergoes DE.
Discovery Rate (CS)	P_a	0.25	The probability (25%) of a cuckoo egg being discovered and replaced by a random solution.
Levy Flight Parameter (CS)	β	1.5	Controls the shape of the Levy distribution step-size.
Crossover Probability (DE)	cr	0.8	The chance (80%) that a parameter in the trial vector is inherited from the mutant vector rather than the target vector.
Scaling Factor (DE)	f	0.5	The differential weight applied during the mutation phase to scale the difference vector.

hand, DE contributes to diversity, maintaining and enhancing the capabilities. Figure 4 shows the strategy used for the research with a proper flow diagram in detail.

3 Results and discussion

3.1 Material removal mechanism in AFM

The surface finish improvement is due to the repeated rubbing of abrasive particles with the surface of the workpiece under extrusion pressure. Abrasive particles remove peaks of the surface in each machining stroke by a micro-cutting and micro-plough mechanism. For a greater number of strokes, the removal process is more uniform, yielding a gradual decrease in surface roughness. Figure 6 shows the impact of optimized selected parameters on the surface roughness. Table 5 shows the experimental results for surface roughness at the mentioned key parameters.

AFM is a non-traditional machining process in which viscoelastic material is rubbed against the surface under extrusion pressure to improve the surface finish [13–19]. The process is suitable for finishing components with complex shape structures having intricate geometries and surfaces that are difficult to finish with traditional methods. This type of structure is generally found in the additive manufacturing process. The effect of stroke length, extrusion pressure, and abrasive concentration at different levels on the signal-to-noise (S/N) ratio is shown in Fig. 5.

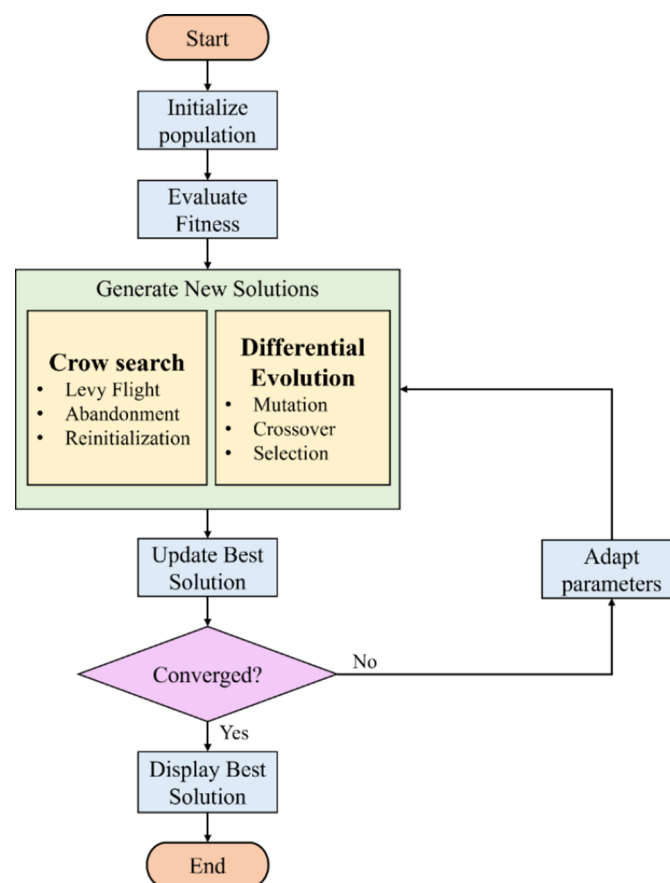


Fig. 4 Flowchart of the Hybrid Crow Search–Differential Evolution (CS–DE) Optimization Process

Table 5 Experimental Results for Surface Roughness

Sr no.	Stroke(no of cycle)	Extrusion pressure(MPa)	Abrasive concentration(wt.%age)	Initial Surface roughness(Ra)	Final Surface roughness(Ra)	%Δ Surface roughness(Ra)	SN ratio
1	50	0.00981	20	10.6	10.2	-3.77	11.527
2	50	0.0490	35	10.9	10.7	-1.83	05.249
3	50	0.0980665	50	10.1	9.30	-7.92	17.975
4	100	0.00981	35	8.6	6.30	-26.7	28.543
5	100	0.0490	50	10.4	10.1	-2.88	09.188
6	100	0.0980665	20	11.0	10.2	-7.27	17.231
7	150	0.00981	50	10.7	9.60	-10.2	20.240
8	150	0.0490	20	10.5	9.80	-6.67	16.483
9	150	0.0980665	35	10.4	10.9	4.81	13.643

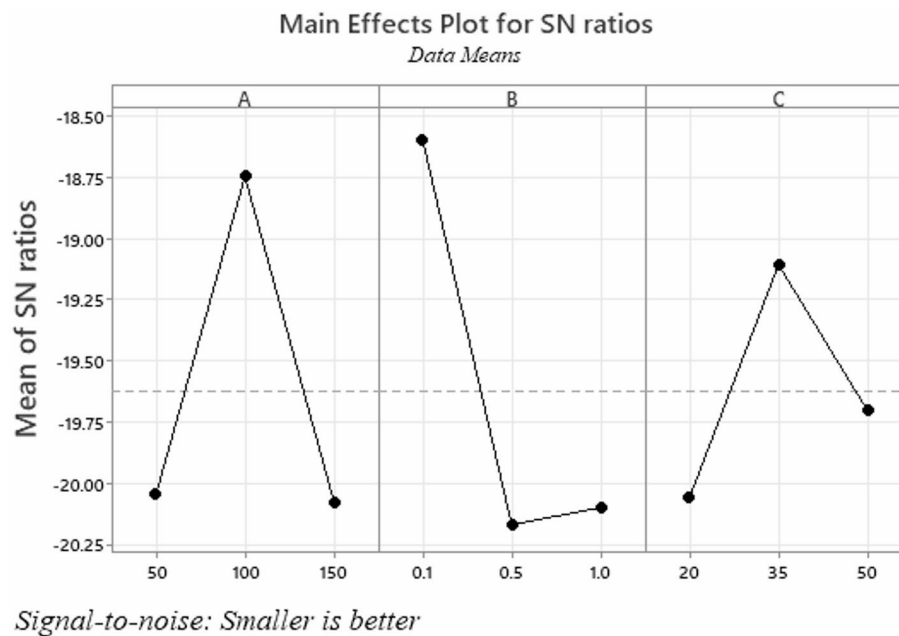


Fig. 5 Effect of stroke length, extrusion pressure, and abrasive concentration at different levels on the signal-to-noise (S/N) ratio

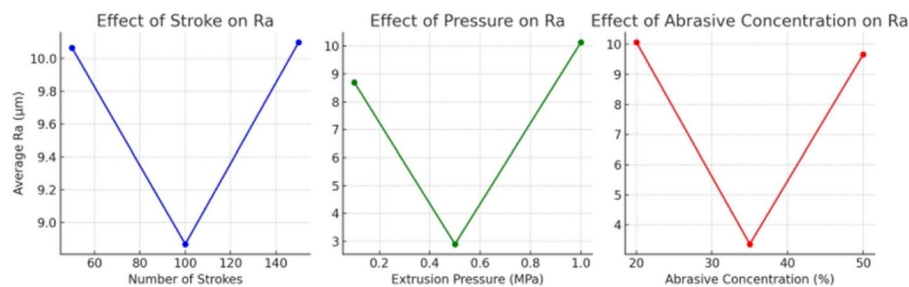


Fig. 6 Impact of optimized parameters (stroke, pressure, and abrasive concentration) on surface roughness

3.2 Effect of stroke on the surface

The changes in surface roughness with abrasive concentration are plotted in Fig. 6a. The average Ra value decreases considerably with the increase of abrasive concentration from 20 to 35%, and the lowest value is obtained at around 35% concentration of the abrasive. This improvement is due to the increase in abrasive content, which means there are more cutting particles in the media, giving more uniform contact with the workpiece surface and improving the material removal process. Anything over 35% concentration leads to a rise in surface roughening again. When abrasive loading is too high, the viscosity of the media may increase, affecting the flowability of the media and abrasive particles' mobility. Consequently, the effectiveness of the micro-cutting action is diminished due to particle crowding and inter-particle interference, which causes non-uniform finishing and increased surface roughness. Thus, a concentration of around 35% is recommended as the most balanced concentration to maximize the cutting efficiency and media flow. Figure 6 shows the impact of the optimized selected parameters on the surface roughness.

3.3 Effect of pressure on the surface

In Fig. 6b, the effect of extrusion pressure on the surface roughness is shown. It can be seen from the above results that the Ra value decreases significantly from about 8.7 μm to about 3.2 μm , which is about 40%, with an increase in extrusion pressure from 0.2 MPa to 0.4 MPa. This improvement is believed to be due to the increase in the normal force acting on the abrasive particles and thus their ability to probe the irregularities more deeply into the surface. Thus, the abrasive particles are capable of conducting an effective micro-cutting operation, which increases material removal and smooths surfaces. But if the pressure is further increased over 0.4 MPa, then surface roughness gradually increases. When the pressure is increased, the particle mobility may be reduced by the excessive compaction of the abrasive media, which causes ploughing instead of controlled cutting. Also, if the contact force is too high, the surface scratches, which will further reduce the surface finish. From these observations, it can be concluded that the optimum extrusion pressure for balancing the cutting efficiency and media flow characteristics is 0.4 MPa.

3.4 Effect of abrasive concentration on the surface

The changes in surface roughness with abrasive concentration are plotted in Fig. 6c. The average Ra value decreases considerably with the increase of abrasive concentration from 20 to 35%, and the lowest value is obtained at around 35% concentration of the abrasive. This improvement is due to the increase in abrasive content, which means there are more cutting particles in the media, giving more uniform contact with the workpiece surface and improving the material removal process. Anything over 35% concentration leads to a rise in surface roughening again. When abrasive loading is too high, the viscosity of the media may increase, affecting the flowability of the media and abrasive particles' mobility. Consequently, the effectiveness of the micro-cutting action is diminished due to particle crowding and inter-particle interference, which causes non-uniform finishing and increased surface roughness. Thus, a concentration of around 35% is recommended as the most balanced concentration to maximize the cutting efficiency and media flow.

3.5 SEM analysis over the surface

SEM images clearly show the evolution of the machined surface. In Fig. 7a, it is clearly visible that machine marks and irregularities are present over the surface. These features were greatly diminished after AFM, and the surface was smoothed and essentially became more uniform. This observation verified to be true that the abrasive media could effectively remove surface peaks by repeated micro-cutting and micro-ploughing actions, which is well in line with the measured reduction in surface roughness. also indicates the comparative surface quality of the specimen before treatment and the surface quality of the specimen after finishing by optimize parameter, which shows that the large voids, valleys, and irregularities are finished. The surface topography profile obtained from the SEM image of the specimen finished with AFM is shown in Fig. 7b. The profile shows a series of ridges and valleys that are characteristic of the FDM process, being the result of the layer deposition pattern. The relatively smooth characteristics of the surface are clearly visible in the smoothed moving-average profile. Figure 7c, d shows the SEM images before and after post-processing, which indicate good abrasion

and material removal from the developed biodegradable abrasive media. The results of the AFM process are promising as the surface shows no discontinuities or surface defects, which indicates that the AFM process is capable of improving the surface quality without compromising the structural integrity of the printed component. Table 6 shows the validation of the research data by means of calculating the percentage error.

3.6 Predictive modelling approaches

Determining the relationships between process parameters and quality to optimize an FDM process through predictive Modelling is essential without having to undertake rigorous experimental testing, and various Modelling methods (including more complex things like long-duration regression models) have therefore been proposed in recent years.

Using the self-adaptive Cuckoo Search algorithm, Chohan et al. [19] created predictive models for surface and dimensional characteristics of vapor-smoothness FDM parts. Fused predictive Modelling with optimization applies the CS algorithm to concurrently determine the best model parameters and process conditions. Combining Modelling and optimization is a very effective approach for the development of the FDM process.

Initial population's best predicted roughness: 6.3178 μm .

Generation 0: Best predicted roughness = 6.234017 μm .

Generation 20: Best predicted roughness = 6.202243 μm .

Generation 40: Best predicted roughness = 6.193624 μm .

Generation 60: Best predicted roughness = 6.193624 μm .

Generation 80: Best predicted roughness = 6.192926 μm .

Generation 100: Best predicted roughness = 6.192333 μm .

Generation 120: Best predicted roughness = 6.192333 μm .

Generation 140: Best predicted roughness = 6.192333 μm .

Vahabli and Rahmati [4] came up with a hybrid estimation technique that blends analytical Modelling with empirical research to determine the distribution of surface roughness in FDM parts. It combined a theoretical insight into the FDM process with the use of experimental data. As a result, the developed models not only describe the basic physics of the process but also incorporate empirical relationships. One can rely on predicting accurately under various operating conditions by using a mixed or hybrid approach to Modelling. This is in stark contrast to using only empirical or only analytical methods, which limit the range of conditions under which predictions may be made. Employed Genetic Algorithm-optimized hybrid learning models for surface roughness prediction [28]. The approach used GA to optimize the architecture and parameters of machine learning models, demonstrating that metaheuristic algorithms can enhance predictive model performance through systematic hyper parameter optimization. Created a hybrid system that integrates Differential Evolution and the Group Method of Data Handling (GMDH) technique for the simulation and prediction of roughness of surfaces in turning processes. Although it is focused on turning, not FDM, this study shows the good performance of hybrid DE-based Modelling approaches for the prediction of surface

Table 6 Validation of Results

Sr no.	Stroke(no of cycle)	Extrusion pressure(MPa)	Abrasive concentration (wt.%age)	Experimental value	Predicted value	%error
1	138	0.009807	33	6.05	6.193624	2.37

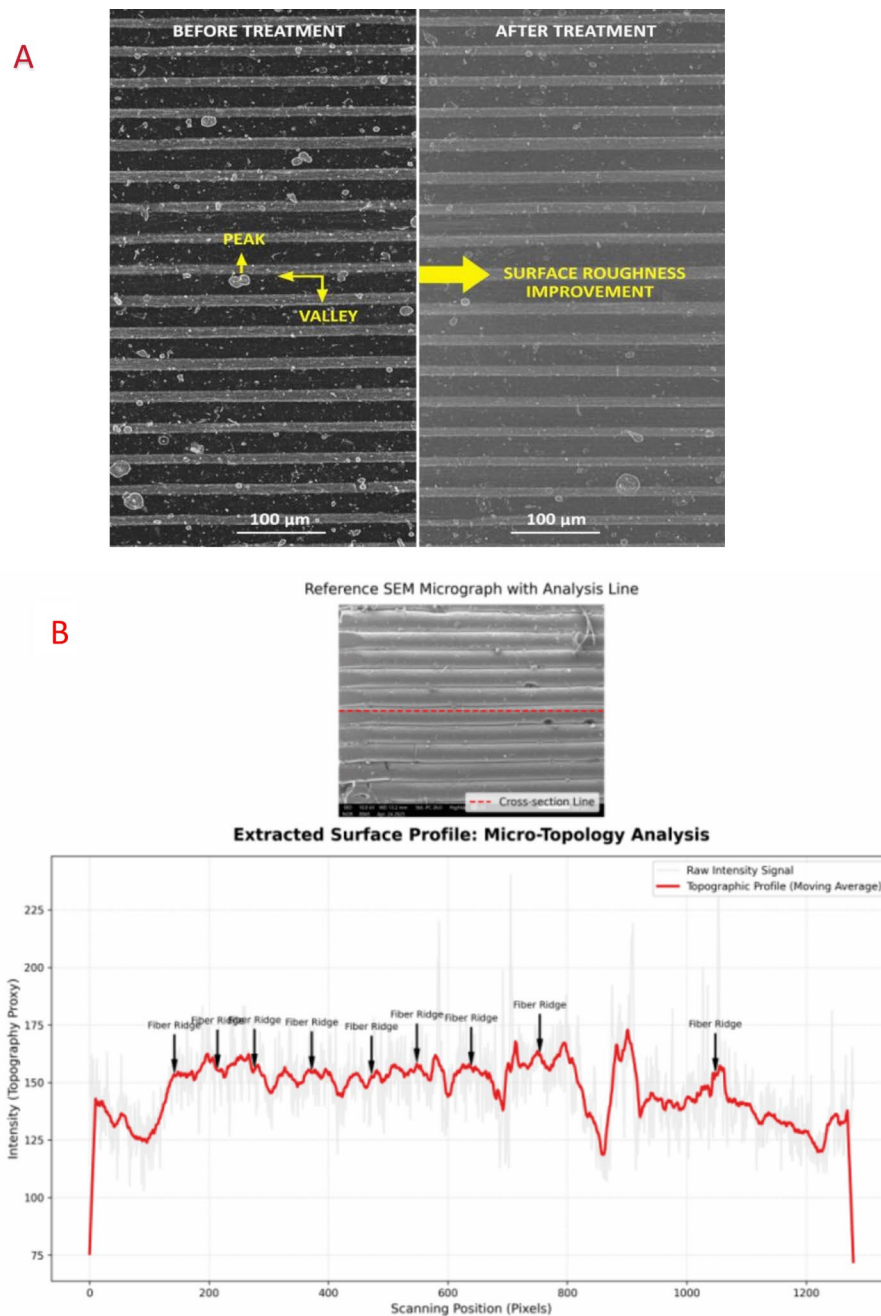


Fig. 7 **a** Comparative impact of optimized parameter before and after; **b** Impact on topology of the part; **c** SEM image before AFM processing; **d** SEM image after post-processing; and **e** Measuring of surface roughness by Tallysurf roughness tester

roughness in manufacturing processes, as shown in Fig. 8. The DE-GMDH algorithm performed better than the regular GMDH and polynomial neural network models, which means that similar hybrid Modelling techniques might also work well in FDM settings [27]. Combining predictive Modelling and optimization algorithms creates a powerful platform for FDM process development. Models provide a means of rapidly evaluating potential parameter combinations during the optimization process. This leads to a significant reduction in computational costs when compared with the evaluation of each candidate solution using simulation-based or experimental methods. The precision

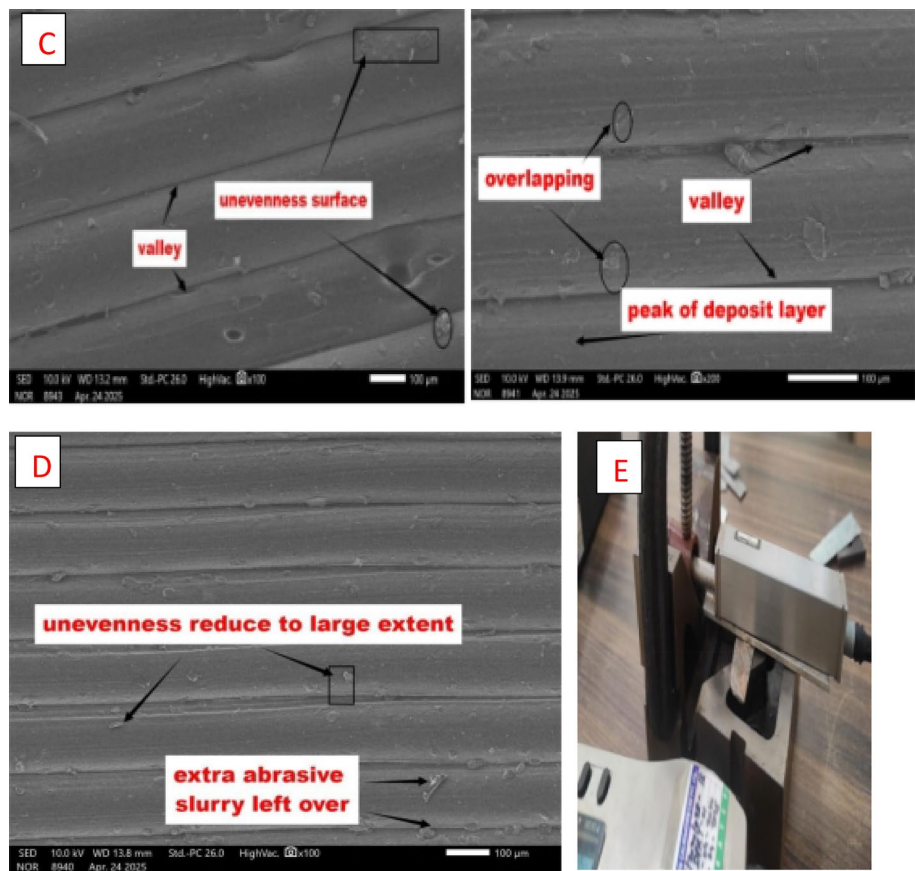


Fig. 7 (continued)

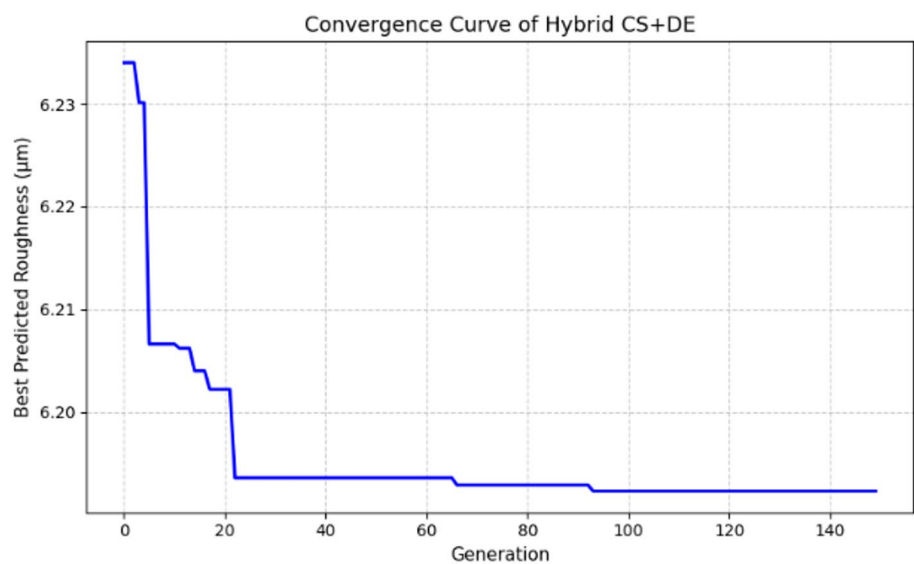


Fig. 8 Convergence curve of hybrid algorithm (Flat regions in the convergence curve indicate that the best solution remained unchanged during those generations, confirming convergence of the hybrid CS-DE algorithm)

of the predictive models chiefly affects the success of the optimization, the development and validation of the models are essential parts of the whole optimization strategy. In the present research, the CSDE hybrid model is used, for which different factors are considered like first initializing the population and the evaluation of fitness, etc.

Chaurasia and Debnath [37] investigate the fabrication of microchannels on biograde Nitinol (NiTi) shape memory alloy (SMA) using micro-electro-discharge (μ -ED) milling, a non-conventional micromachining technique suited for ultra-hard and exotic materials. The optimized process achieved a maximum MRR of 1.267×10^{-6} kg/s, a minimum dimensional deviation of 16.47 μ m, and a minimum surface roughness of Ra 0.387 μ m, demonstrating the effectiveness of this approach for producing biocompatible microstructures relevant to biomedical device applications that provide suitable results, but in the present study material method and process considerations are totally different; therefore, the results are different and show high variations [39–41].

3.7 Mathematical modelling and optimization

Through mathematical Modelling of the AFM process, surface finish or optimization of the process parameters for the outcome of the surface can be easily predicted. Dhull et al. [14] carried out experimentation on the mathematical model for optimization of AFM process parameters, taking the basic parameters that play an important role in the AFM process, like abrasive concentration and pressure, and performed the optimization over it. Hashmi et al. [19] worked on a mathematical model which integrates with numerical simulation, that gives insight into abrasive flow finishing processes [19]. The work largely depends upon the material removal mechanism, and their research goes beyond the empirical relation of the parameter, so the process gives better and deeper process behavior. Gupta et al. [15] have researched the impact of surface roughness and how it is optimized by changing process parameters in AFM. The study proved that a systematic optimization of parameters could result in an enormous improvement in AFM performances than the random testing methods. Kumar et al. [17] focused on process parameter optimization of material surface roughness in AFM to get the maximum effect. Through design of experiments and statistical analysis, the study helped to identify the optimal parameter setting, thus proving that structured optimization methodologies are a very effective means of performance improvement. Kumar et al. [21] also describe Electro-Chemical AFM (ECAFM), which is a combination of AFM and electro-chemical, where the latter is used as the abrasive material. It used the Taguchi method as a tool for optimization of their parameters and also shows how the hybrid model combined with parameters works better and provides the best and most desired solution for the same. Diyaley et al. [38] analyzed different metaheuristics for parametric optimization of abrasive water-jet machining, which is a related abrasive-based process. Perhaps similar techniques could be employed to solve the problem of AFM optimization. Thus, combining mathematical models and metaheuristic optimization algorithms allows us to create the potential framework for AFM parameter optimization [42–46]. Once the model is established, the evaluation of a proposed parameter set can be done swiftly, even with a view to finding optimum connection settings. Metaheuristic algorithms enhance the search in parameter space, as well as the effective utilization of runtime parameters, turning up settings that are best suited to a given test or process. It is especially worthwhile for AFM optimization when experimental evaluations necessitate

a huge amount of time, cost, and effort for every single or multiple configurations of the accepted test specimens.

A polynomial regression model integrated with the optimization algorithm was employed to determine the optimal process parameters for minimizing the surface roughness during AFM. The optimization results predicted that the minimum surface roughness could be achieved at a stroke count of 138.1284 cycles, an extrusion pressure of 0.0098 MPa, and an abrasive concentration of 32.8423 wt%. Under these optimum conditions, the predicted minimum surface roughness was found to be 6.192333 μm .

The developed polynomial regression model demonstrated excellent predictive capability. The statistical performance indicators revealed an R^2 value of 0.9831, indicating that approximately 98.31% of the variation in surface roughness is explained by the developed model. Furthermore, the model exhibited a very low Mean Squared Error (MSE) of 0.002322 and Mean Absolute Error (MAE) of 0.03948, confirming the high accuracy and reliability of the prediction model.

The relationship between process parameters and surface roughness was represented by a third-order polynomial regression equation. The third-order polynomial regression was chosen because, according to the model performance measures (R^2 , MAE, MSE, and RMSE), it offered the best predictive performance among the models assessed in MATLAB.

$$\begin{aligned} \text{Surface Roughness} = & 25.516942 + (-0.451756 \times \text{Stroke}) + (-8.078400 \times \text{Pressure}) \\ & + (0.063974 \times \text{Concentration}) + (0.002529 \times \text{Stroke}^2) \\ & + (0.081860 \times \text{Stroke} \times \text{Pressure}) + (0.006850 \times \text{Stroke} \times \text{Concentration}) \\ & + (-4.077563 \times \text{Pressure}^2) + (0.891985 \times \text{Pressure} \times \text{Concentration}) \\ & + (-0.020404 \times \text{Concentration}^2) + (-0.000004 \times \text{Stroke}^3) \\ & + (0.000190 \times \text{Stroke}^2 \times \text{Pressure}) + (-0.000034 \times \text{Stroke}^2 \times \text{Concentration}) \\ & + (0.051270 \times \text{Stroke} \times \text{Pressure}^2) + (-0.004972 \times \text{Stroke} \times \text{Pressure} \times \text{Concentration}) \\ & + (0.000056 \times \text{Stroke} \times \text{Concentration}^2) + (-1.377690 \times \text{Pressure}^3) \\ & + (-0.151514 \times \text{Pressure}^2 \times \text{Concentration}) \\ & + (-0.004431 \times \text{Pressure} \times \text{Concentration}^2) + (0.000171 \times \text{Concentration}^3) \end{aligned}$$

To validate the optimization results, an experimental trial was conducted near the predicted optimum region as shown in Table 6. The experimental surface roughness obtained at 138 cycles, 0.009807 MPa pressure, and 33 wt% abrasive concentrations was 6.05 μm , while the regression model predicted a value of 6.193624 μm . The percentage error between the experimental and predicted values was only 2.37%, demonstrating strong agreement between the model predictions and experimental observations.

The low prediction error confirms the robustness of the developed polynomial regression model and validates the effectiveness of the optimization algorithm in identifying suitable AFM process parameters for achieving improved surface finish. The results indicate that lower extrusion pressure combined with a moderate abrasive concentration and optimized stroke count significantly contributes to reducing surface roughness. Consequently, the proposed optimization methodology can be effectively utilized for process parameter selection and performance enhancement in AFM applications.

The error value of 2.37% indicates excellent model accuracy and demonstrates that the developed regression-based optimization framework can reliably predict surface roughness within the investigated process domain. Such close agreement between predicted

and experimental values further strengthens the applicability of the proposed hybrid optimization approach for industrial AFM process optimization.

4 Future implications

An extensive comparative research of the proposed CS-DE algorithm and other state-of-the-art optimization methods is an important research direction. These studies will allow for the benchmarking of the optimization accuracy, convergence rate, computational cost, and stability of the proposed algorithm, which will help enhance the knowledge of the algorithm's capabilities and limitations. The present study mainly aims at the evaluation of machining performance of the fabricated biodegradable abrasive media, but comprehensive physicochemical characterization of the media is a promising research trajectory. In the future, the viscosity, the Rheological behavior, the particle size distribution, the stability of the media, and the flow behavior of the media should be measured so that a deeper insight into the transport behavior, abrasive dispersion, and long-term stability of the media during the machining process can be gained. This characterization will further validate the performance of the proposed biodegradable abrasive media and help to optimize it for industrial AFM applications.

5 Conclusion

Additive manufacturing is the most suitable process to manufacture various parts, the drawback which the process has can be removed or diminished by using various empirical as well as mathematical computational methods, if the combination process is used then it gives multi-level benefits. As per the synthesis of current study, investigate the surface finish of PLA/Cu reinforced FDM made tubular specimens using AFM with biodegradable media containing paper pulp, peanut oil and stainless steel 316 L particles. The experimental investigation was conducted to evaluate the major parameters, extrusion pressure, no. of strokes, and abrasive concentration of AFM for surface finish. The results demonstrate that all three parameters contribute to the surface quality of the specimen. The extrusion pressure plays a more significant role in the surface quality. The hybrid optimization method Cuckoo Search–Differential Evolution (CS-DE) successfully identifies the optimum solution at 138 strokes, 0.00981 MPa extrusion pressure, and 33% abrasive concentration of AFM processing conditions. Under this condition, a surface roughness found to be $6.05 \mu\text{m}$ was achieved. Representing the surface roughness of -26.7 is improved from the printed FDM part, which was found to be 8.6 in the initial stage. The key scientific achievement of this work is the development of an environmentally friendly abrasive media and the successful implementation of a hybrid CS-DE optimization framework for the optimization of AFM parameters. The coefficient of determination (R^2) was determined as 98.31%, which means the model explained 98.31% of the variance in surface roughness. The model is able to explain the overall trend of the process. The value of R^2 , which is quite good, indicates that there are linear relationships that are affecting the response. The profile image of the surface profile showed a fairly uniform distribution of ridges and lower amplitude of surface roughness variations at the optimized process parameters, which proved the effectiveness of the hybrid CS-DE optimized AFM parameters in reducing the surface roughness. The SEM profile confirms the improvement in surface finish observed in the experiment and further confirms the optimization results. The methodology suggested in the proposed approach

is a very effective and environmentally friendly method for enhancing the surface characteristics of the polymer-metal composite parts produced by additive manufacturing techniques, which are used for engineering applications.

Author contributions

Conceptualization: Vinay Shah, Satbir Singh Sehgal, Jasgurpreet Singh Chohan; Methodology: Vinay Shah, Satbir Singh Sehgal, Jasgurpreet Singh Chohan, Sachin Kalsi, Bonsa Regassa Hunde; Writing - Original Draft: Vinay Shah, Satbir Singh Sehgal, Jasgurpreet Singh Chohan, Sachin Kalsi, Bonsa Regassa Hunde; Writing - Review & Editing: Sachin Kalsi, Bonsa Regassa Hunde; Supervision: Satbir Singh Sehgal, Jasgurpreet Singh Chohan; Data Curation: Vinay Shah, Bonsa Regassa Hunde.

Funding

Not applicable.

Data availability

The data supporting the findings of this study are available on request from corresponding author.

Declarations

Conflict of interest

The authors declare no conflict of interest.

Ethical approval

This study did not involve human participants or live animals. Therefore, no ethical approval is required for the study.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Received: 14 May 2026 / Accepted: 3 September 2026



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